



Depleting high-abundant and enriching low-abundant proteins in human serum: An evaluation of sample preparation methods using magnetic nanoparticle, chemical depletion and immunoaffinity techniques

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ABSTRACT

The efficiency of three different depletion methods to remove the most abundant proteins, enriching those human serum proteins with low abundance is checked to make more efficient the search and discovery of biomarkers. These methods utilize magnetic nanoparticles (MNPs), chemical reagents (sequential application of dithiothreitol and acetonitrile, DTT/ACN), and commercial apparatus based on immunoaffinity (ProteoMiner, PM). The comparison between methods shows significant removal of abundant protein, remaining in the supernatant at concentrations of 4.6 ± 0.2 , 3.6 ± 0.1 , and $3.3 \pm 0.2 \mu\text{g mL}^{-1}$ ($n=3$) for MNPs, DTT/ACN and PM respectively, from a total protein content of $54 \mu\text{g mL}^{-1}$. Using GeLC-MS/MS analysis, MNPs depletion shows good efficiency in removing high molecular weight proteins ($> 80 \text{ kDa}$). Due to the synergic effect between the reagents DTT and ACN, DTT/ACN-based depletion offers good performance in the depletion of thiol-rich proteins, such as albumin and transferrin (DTT action), as well as of high molecular weight proteins (ACN action). Furthermore, PM equalization confirms its efficiency in concentrating low-abundant proteins, decreasing the dynamic range of protein levels in human serum. Direct comparison between the treatments reveals 72 proteins identified when using MNP depletion (43 of them exclusively by this method), but only 20 proteins using DTT/ACN (seven exclusively by this method). Additionally, after PM treatment 30 proteins were identified, seven exclusively by this method. Thus, MNPs and DTT/ACN depletion can be simple, quick, cheap, and robust alternatives for immunochemistry-based protein depletion, providing a potential strategy in the search for disease biomarkers.

1. Introduction

Human blood (serum and plasma) has been extensively studied to identify and quantify proteins, focusing on biomarkers that may help in the early diagnosis, prognosis and progression of diseases [1–3]. However, the complexity of such samples, in addition to the wide dynamic range of proteins, imposes many analytical challenges [4,5]. At minimum, tens of thousands of different proteins may be present in serum samples, with a concentration range of such proteins ranging to 10 orders of magnitude [6]. Interestingly, just 22 proteins account for more than 99% of the total protein content in serum, comprising the most abundant proteins: albumin, myoglobin, immunoglobulins, transferrin and haptoglobin [6,7]. However, this group of proteins may interfere in the detection of those low abundance proteins that are

currently considered as potential biological markers of disease [8,9]. In spite of the great advances in analytical instrumentation, such as the advent of high-resolution mass spectrometry, an accurate sample preparation protocol remains fundamental for solving/minimizing the complexity of the samples, thus contributing to overcoming the complexity of the serum proteome [10]. Some strategies have been employed in this context [1,3,8,10], such as immune-depletion columns for removing those high abundance proteins [5], which present some limitations, such as low sample throughput, non-specific loss of albumin-proteins and species specificity issues [10]. On the other hand, non-chromatographic approaches have also been proposed as cost-effective and quick methods for simplifying the proteomes, such as sequential use of acetonitrile (ACN) [3] and dithiothreitol (DTT) [10], the use of commercial kit such as ProteoMiner (PM) [1], and, most

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recently, the use of coated magnetic nanoparticles (MNPs) [8], focusing on biomarker discovery.

Thus, the aim of this study is to evaluate three different methods for simplifying the complex proteome of serum samples, removing the most abundant proteins and enriching those of low abundance. Such methods use 1) crude MNPs, 2) DTT followed by protein depletion with ACN, and 3) protein equalization with the commercial ProteoMiner® apparatus. In fact, the first and second methods are new, and do not yet possess a recognized protocol; few references are found in the literature with similar application [3,8]. The third is in current use, and is used as a reference.

2. Experimental

2.1. Chemicals and reagents

All reagents were LC or electrophoresis grade, except in the synthesis of MNPs. Dithiothreitol (DTT), iodoacetamide (IAA), urea, thiourea, 3-[(3-Cholamidopropyl) dimethylammonio]-1-propane-sulfonate (CHAPS), glycerol, sodium dodecyl sulfate (SDS), Tris-base, glycine, N,N,N',N'-tetramethylethylenediamine (TEMED), acrylamide/bis-acrylamide, bromophenol blue, and mineral oil were purchased from GE Healthcare (Uppsala, Sweden). Acetonitrile, ethanol, hydrochloric acid (HCl), formic acid, β -mercaptoethanol, trifluoroacetic acid (TFA), ammonium persulfate (APS), coomassie brilliant blue G-250 (CBB) were obtained from JT Baker (Mex, Mexico). Iron (III) chloride hexahydrate, ferrous sulfate heptahydrate and concentrated ammonia solution (NH_3 25%, w/w) were obtained from Synth (SP, Brazil). Phosphate buffered saline (PBS) pH 7.4 was purchased from Sigma Aldrich (MO, USA), and the ProteoMiner® Protein Enrichment Small-Capacity (PM) kit from Bio-Rad (CA, USA). Deionized water (resistivity $> 18.0 \text{ M}\Omega \text{ cm}$) was produced using a Millipore (France) water purification system. Trypsin gold was obtained from Promega (Wi, USA), and molecular weight standards for gel electrophoresis acquired from Bio-Rad (CA, USA).

2.2. Serum sample

Human serum samples were obtained from anonymous donors at Clinical Hospital, University of Campinas, Brazil. The local Ethics Committee (protocol number 775/2010) approved this study. Blood was drawn into vacutainer tubes, immediately placed in ice, allowed to clot for at least 30 min, and centrifuged at $1500g$ for 15 min. The serum obtained was aliquoted, transferred into LoBind tubes containing 0.01% (w/v) sodium azide, and stored at -80°C until assayed.

2.3. Synthesis of Fe_3O_4 magnetic nanoparticles

The synthesis of magnetic nanoparticles was performed according to a previously described procedure [11]. Briefly, 6 mL of concentrated ammonium hydroxide and 4 mL of water saturated with N_2 were mixed in a round-bottom flask under an inert atmosphere. Subsequently, in another vessel, 1 g of $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and 0.25–0.5 g of $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ were dissolved in 10 mL of water saturated with N_2 . After mixing both solutions, the system was kept under constant stirring for 80 min at 80°C . The final reaction product was washed three times with deionized water, and magnetically separated for 24 h at room temperature.

2.4. MNPs characterization

SEM images were acquired with a scanning electron microscope JEOL JSM-360 - LV, operating at 20 kV with secondary electron image. Samples were analyzed on a sample support of Cu-Zn metallized with a 10 nm gold layer (sputter coating Bal-tec MED020). X-ray diffraction (XRD) pattern was obtained by a Shimadzu XRD600 using Cu-K α

radiation, 40 kV voltage, 30 mA current, and graphite monochromator; measurements were made from 10 to 70°C (2 θ). Nitrogen adsorption isotherm measurements were carried out using a Micrometrics ASAP 2020. The sample was degassed at 180°C for 6 h prior to measurement. For thermal analysis (TG/dTG), approximately 10 mg of synthesized solid were placed in an alumina sample holder for analysis in a TGA/DTA Setaram, SETSYS 16/18 model, coupled to a Thermostat mass spectrometer (Pfeiffer). Samples were analyzed under oxygen stream (16 mL min^{-1}), from 20 to 1000°C , at a heating rate of $10^\circ\text{C min}^{-1}$. DLS analysis was performed at 25°C , with a Mastersizer 2000 from Malvern. Initially, a solid sample of nanoparticles was dispersed in water and submitted to ultrasound for 30 min. After sonication, 2 mL of the suspension was transferred to a quartz cuvette and measurement was made with the detector at a fixed scattering angle of 173° , using a He-Ne laser and Zetasizer software.

2.5. Sample preparation

2.5.1. Protein depletion using magnetic nanoparticles (MNPs)

High-abundance proteins depletion by crude MNPs from serum samples was investigated batch-wise, and the effects of nanoparticle size, incubation time, MNP/protein ratios, and pH of the medium on the depletion process were evaluated. For this study, shaking at 300 rpm and an incubation temperature of 25°C were predefined experimental conditions. The effect of sample pH on the depletion procedure was evaluated from pH 5.5 to 8.5 using Tris-HCl solution. A series of serum sample aliquots were used for checking the influence on protein present during the depletion process. Volumes of serum (at $54 \mu\text{g mL}^{-1}$ total protein concentration) ranging from 25 to 90 μL were diluted to a final volume of 100 μL in PBS (2% v/v, pH 7.4) or Tris-HCl (0.1 mol L^{-1} , pH 5.5 or 8.5), and mixed with MNPs (5 μg) in order to obtain the following MNP/protein ratios: 1:2, 1:5, 1:10. Additionally, the extraction time (from 30 to 120 min), and MNP size (30 and 70 nm) were also investigated. After the depletion, the supernatant and solid phase, which contained the adsorbed major protein on the MNP surface, were then magnetically separated. The supernatant was transferred to a clean LoBind tube, and evaporated to dryness in a vacuum concentrator centrifuge without heating. The sample was reconstituted in 100 μL of a solution containing urea (8 mol L^{-1}), CHAPS (2% m/v), thiourea (3 mol L^{-1}), and stored at -20°C until gel electrophoresis analysis.

2.5.2. Sequential protein depletion using DTT and ACN

For this task, two protocols, previously described [3] using DTT and ACN, were performed in sequence. Briefly, three aliquots of 20 μL of human serum were mixed with 2.0 μL DTT [500 mmol L^{-1} , solubilized in AMBIC (12.5 mmol L^{-1})] and vortexed. The sample was incubated for 60 min, centrifuged at $13,000g$ for 40 min, and the supernatants transferred to a LoBind microcentrifuge tube (Eppendorf, Cambridge, UK). Each aliquot was diluted with 45 μL of water and vortexed, then added to 85 μL of ACN and sonicated for 10 min in an ultrasonic bath. Samples were quickly vortexed and further sonicated for 10 min. The protein precipitate was pelleted by centrifugation at $13,000g$ for 10 min at room temperature. The supernatants obtained from those three aliquots were pooled in a clean LoBind tube, and evaporated to dryness in a vacuum concentrator centrifuge without heating. The sample was reconstituted in 100 μL of a solution containing 8 mol L^{-1} urea, 2% m/v CHAPS, 3 mol L^{-1} thiourea, and stored at -20°C until gel electrophoresis analysis.

2.5.3. Protein equalization using ProteoMiner® (PM)

The protein enrichment and depletion of serum samples with the PM apparatus was carried out according to the manufacturer's instructions. Briefly, PM spin columns were first washed with water (2 $\times$ 5 min), and, then, with washing buffer (150 mmol L^{-1} NaCl, 10 mmol L^{-1} NaH_2PO_4 , pH 7.4) (2 $\times$ 5 min). Four aliquots of human

serum (200 μL) were then applied to the column and incubated for 2 h. Afterwards, the column was washed with buffer (3 $\times$ 5 min). Protein elution was performed three times for 15 min with 100 μL rehydrated elution reagent (containing urea, CHAPS and acetic acid). Eluted fractions were adjusted to neutral pH with a 1 mol L^{-1} solution of Tris and stored at $-20\text{ }^{\circ}\text{C}$.

2.6. Protein quantification

After depletion procedure, the protein concentration was determined using the 2-D Quant protein assay kit (GE, Healthcare), according to the manufacturer's instructions [12]. Briefly, an analytical curve for bovine serum albumin (BSA) was built up from 0.0 to 50.0 $\mu\text{g } \mu\text{L}^{-1}$ linear range (in duplicate). Aliquots of 3 μL of each sample (in triplicate) were transferred to a LoBind tube, and a specific solution (see manufacturer's instruction) was added to each tube for protein precipitation. The tubes were then centrifuged for 5 min at 10,000*g*, the supernatant removed and copper solution and de-ionized water added to each tube. After a short mixing to dissolve the precipitated protein, the color reagent (see manufacturer's instruction) was added, and the mixture incubated at room temperature for 15 min. The solutions were then transferred to 96-well microplate and the absorbance measured at 486 nm. The calibration curve was then used to determine the protein concentration in the samples. Each experiment was performed in triplicate for quality control and statistical purposes. The following equation describes the depletion efficiency (E):

$$E = (C_e/C_0) \times 100\%$$

where C_e is the equilibrium protein concentration ($\mu\text{g } \mu\text{L}^{-1}$), and C_0 is the initial protein concentration ($\mu\text{g } \mu\text{L}^{-1}$).

2.7. Gel electrophoresis

Amounts of 30 μg protein of crude serum sample or depleted serum sample were mixed with sample buffer (10% *w/v* SDS, Tris-base 40 mmol L^{-1} , pH 6.8, 50% *v/v* glycerol, 0.1% *v/v* bromophenol blue, 10% *v/v* β -mercaptoethanol) in a proportion of 1:1. Heating at 98 $^{\circ}\text{C}$ for 3 min promoted the sample denaturation, and they were then loaded onto 12% (*w/w*) polyacrylamide gels of 1 mm thickness. The electrophoretic separation was performed as follows: 200 V for 50 min, and the gel stained with CBB.

2.8. Stain and image analysis

CBB stain solution was prepared by dissolving 1 g of CBB G-250 in 100 mL of a mixture of 50% *v/v* methanol, 10% *v/v* acetic acid and 40% *v/v* water. After running, the gels were incubated in the staining solution for 1 h at room temperature, and then the solution removed by washing the gel with water. Gel destaining was performed using a solution containing 40% *v/v* methanol, 10% *v/v* acetic acid and 50% *v/v* water, for 24 h at room temperature. The gels were then rinsed with water and incubated for at least 1 h. Images were acquired using an ImageScanner (GE Healthcare).

2.9. In-gel protein digestion

Protein bands were manually excised from the CBB-stained gels and subjected to in-gel trypsin digestion, as described previously [13]. Briefly, the bands excised from the gels were placed onto a micro-SPE plate containing peptide affinity resin, and the digestion and vacuum elution protocols were performed according to the manufacturer's recommendations using the Montage® In-GEL digestZP kit (Millipore, Bedford, USA). First, each protein spot was destained with acetonitrile, and trypsin added. For in-gel trypsin digestion, each spot was incubated overnight with 166 ng of trypsin at 37 $^{\circ}\text{C}$. The peptides were then extracted from the resin, twice, using TFA (0.1%, *v/v*) in

ACN (50%, *v/v*). The vacuum elution was performed using a Multiscreen® Vacuum Manifold (Millipore) system.

2.10. NanoLC-MS/MS analysis

After trypsin digestion, the peptides obtained were dried and resuspended in formic acid (0.1%, *v/v*). The resulting peptide mixture was analyzed by LC-MS/MS using a nanoAcquity UPLC (Waters Co., Manchester, UK), coupled to a Waters Synapt HDMS (Waters Co.) mass spectrometer, and equipped with a BEH C18 column (100 $\mu\text{m} \times 100\text{ mm}$, Acquity Waters). Chromatographic separations were performed at gradient mode from acetonitrile (2–90%) in formic acid (0.1%, *v/v*) with 1.0 $\mu\text{L min}^{-1}$ flow rate for 20 min. The mass spectrometer was handled by Data Dependent Analysis (DDA) software, which acquires one spectrum per second. When detecting multi-charged species, the three most intense species were fragmented in the collision cell (collision energy set according to *m/z* precursor and charge). The spectra acquisition was performed using MassLynx v.4.1 software.

Files obtained from the acquired mass spectra were converted to a peak list format using Mascot Distiller (Matrix Science), and searched against the National Center for Biotechnological Information (NCBI) and Swiss-Prot protein databases. The Mascot Server 2.3 MS/MS search engine was used with the following parameters for protein identification: carbamidomethylation of cysteine as the fixed modification, oxidation of methionine as the variable modification, one missed trypsin cleavage, and a tolerance of $\pm 0.1\text{ Da}$ for the precursor and fragment ions. The significance threshold was set at $p < 0.05$, which corresponds to a minimum score 30.

3. Results and discussion

3.1. Protein depletion/equalization efficiency

After determining the optimal condition for depletion using MNPs, the reproducibility of each method tested in this study, as well as its efficiency in reducing the protein content of serum samples, were assessed. To this end, those three methods already described were then used, employing a series of aliquots of the same serum sample. Once the aliquots were depleted or equalized as described in the experimental section, the remaining protein content was calculated in $\mu\text{g } \mu\text{L}^{-1}$. Thus, the supernatant content was 4.6 ± 0.2 , 3.6 ± 0.1 , and $3.3 \pm 0.2\text{ } \mu\text{g } \mu\text{L}^{-1}$ ($n=3$) for the MNPs, DTT/ACN and PM methods respectively (see Fig. 1, panel A). Student's *t*-test at the significance level of $p < 0.05$ was used for checking the differences between depletion methods.

3.2. Characterization of the Fe_3O_4 MNPs

Fe_3O_4 MNPs were synthesized by a co-precipitation method, using two different masses of Fe(II) ions (0.25 and 0.50 g) to obtain particles of different sizes. Fig. 2 shows the particle size distributions obtained by DLS and SEM images for the materials synthesized from 0.25 g (Fig. 2, panel A) and 0.50 g (Fig. 2, panel B) Fe(II) ion solutions. SEM images showed no morphological differences between the particles, both being spherical, with similar degree of agglomeration, although sizes were different. DLS measurements clearly indicate average sizes of 30 nm and 70 nm for particles obtained from 0.25 and 0.50 g of Fe(II) ions respectively, demonstrating that particle size is lower as the ratio Fe(II)/Fe(III) ions in the reaction mixture is lower. In addition, surface areas for 30 nm and 70 nm MNPs were determined as 234 and 170 $\text{m}^2\text{ g}^{-1}$ respectively. The TG/dTG curves for Fe_3O_4 MNPs are shown in Fig. 2 (panel C), showing the thermal stability of MNPs and their composition. The weight loss in the 80–140 $^{\circ}\text{C}$ range can be assigned to water thermo-desorption from the sample. Additionally, during the middle temperature range 200–300 $^{\circ}\text{C}$, the dTG curves of

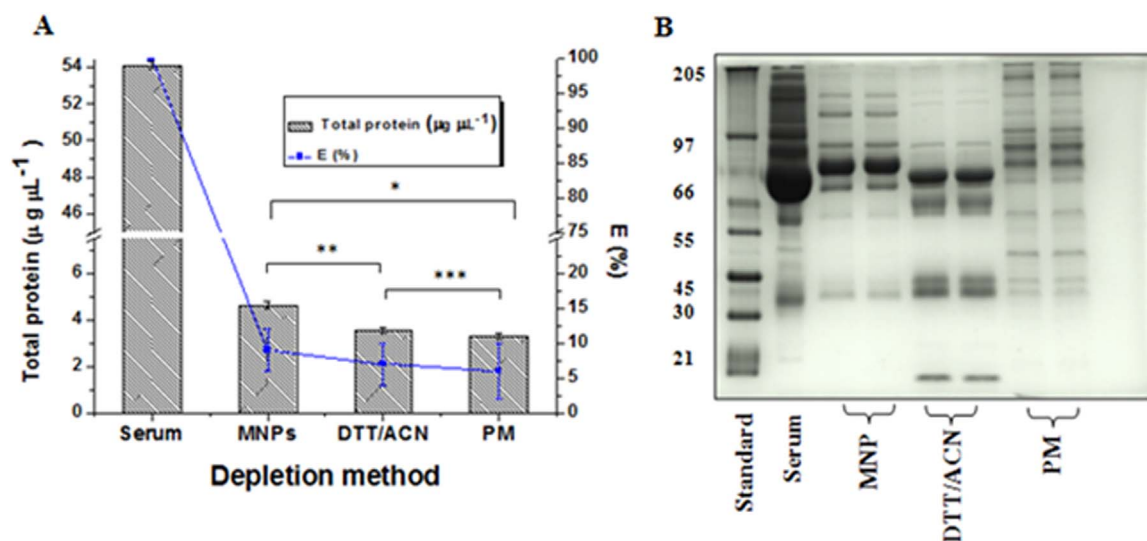


Fig. 1. Comparison of the depletion methods using MNPs, DTT/ACN and PM. (A) Protein concentration ($\mu\text{g } \mu\text{L}^{-1}$) measured with 2-D Quant kit (GE). Results obtained from three independent depletion experiments. The square curve represents the protein recovery (%) remaining after the depletion. Statistical analysis were performed using *t*-test and considered significant for a * $p < 0.001$, ** $p < 0.05$, and *** $p > 0.05$. (B) Representative 1-DE gel running of the samples, in duplicate. A fixed amount of protein (30 μg) was loaded per lane for all the samples.

Fe_3O_4 MNPs exhibit a weight loss related to the formation of $\gamma\text{-Fe}_2\text{O}_3$ (maghemite). The final weight loss occurs above 682°C , and it is attributed to the phase transition from $\gamma\text{-Fe}_2\text{O}_3$ to FeO . Fig. 2 (panel D) shows a typical XRD pattern of a Fe_3O_4 MNPs sample, where the diffraction peaks with 2θ at 30.2° , 35.5° , 43.1° , 53.4° , 57.0° and 62.6° were observed, which indicated the formation of a crystalline structure of inverse spinel type, characteristic of magnetite (JCPDS card No. 75-1609).

3.3. Optimization of parameters for high-abundance protein depletion using MNPs

In recent years, magnetic nanoparticles have attracting more attention as an important strategy for protein separation of biological samples, because they provide simple, cheap and robust applications in the discovery of human disease biomarkers [14]. Although convenient and quick, such strategies present a great number of variables that may influence the efficiency of protein adsorption on the MNPs surface [15,16]. For this reason, four parameters were evaluated in triplicate for quality control and statistical purposes: (i) MNPs size; (ii) incubation time; (iii) MNP/protein ratios; and (iv) pH of the medium. Shaking and incubation temperature were previously defined as 300 rpm and 25°C , respectively.

3.3.1. Influence of MNP size

Recent studies have reported that MNP size plays an important role in the protein adsorption to their surface, influencing the protein abundance composition of a biological fluid, such as serum [17,18]. Thus, for demonstrating the influence of MNPs size in the high-abundance protein depletion process, two different sizes (30 and 70 nm) of MNPs were evaluated. The experiment was performed using an initial ratio of 1:2 (MNP/protein), a pH of 7.4, and shaking for 30 min. After 30 min of incubation, the supernatant and MNPs containing the adsorbed proteins were then magnetically separated. The remaining protein in the supernatant was quantified for evaluating the depletion efficiency. The results are presented in Fig. 3 (panel A), which shows that both nanoparticles (30 and 70 nm) present good efficiency in removing serum sample protein compared to crude serum. Nevertheless, although the 70 nm MNPs present lower surface area, they provided, curiously, better results in terms of protein depletion, achieving a protein recovery of ca. $4 \mu\text{g } \mu\text{L}^{-1}$ (Fig. 3, panel A, graphic

bar), while 30 nm MNPs achieved recoveries of ca. $7 \mu\text{g } \mu\text{L}^{-1}$ (Fig. 3, panel A, graphic bar). These results may be explained in terms of agglomeration of the MNPs. In other word, we hypothesized that 30 nm MNPs may have suffered a rapid *in situ* agglomeration, influencing the adsorption between proteins and 30 nm MNPs. Nevertheless, to 70 nm MNPs, such *in situ* agglomeration was minimal or nonexistent. Thus, this fact indicating that MNPs aggregation also affects the effective interaction between proteins and MNPs. In addition, our results support those findings by Hu et al. [18] suggesting that at the same mass concentration, larger nanoparticles are significantly better than smaller in the depletion of high abundance protein. Thus, for further experiments, 70 nm MNPs were utilized for removing high abundance protein from serum samples.

3.3.2. Influence of the MNP/protein ratio

Protein concentration is another important parameter which may affects the capacity and kinetics of protein adsorption [15]. To investigate the influence of the MNP/protein ratio on the depletion process, a series of aliquots of serum sample were mixed with MNPs (70 nm), at MNP/protein ratios of 1:2, 1:5, and 1:10 (see Section 2.5.1). Fig. 3, panel B, shows a decrease in depletion efficiency, with an increase in protein concentration. This observation is explained by the saturation of the system. In fact, maintaining the amount of adsorbent (i.e. MNPs) constant and increasing the protein concentration, would be expected to lead to a decrease of available adsorption sites, reducing the efficiency of protein removal [14]. As a compromise between MNPs and protein extraction, the 1:2 ratio (MNP/protein) was then selected for future experiments.

3.3.3. Influence of incubation time

The effect of incubation time on the protein depletion process was evaluated by changing the shaking time (30, 60, and 120 min). Fig. 3, panel C, shows no significant change in the depletion efficiency with shaking time evaluated. This result is justified based on the presence of many MNPs surface adsorption sites available to the proteins at the beginning of the extraction process, and which reduce during extraction, making fewer sites available to the proteins, and gradually slowing the entire extraction process until it reaches equilibrium [14]. Then, 30 min shaking was enough to guarantee adequate protein adsorption on the MNPs surface, and it was selected as the optimum incubation time.

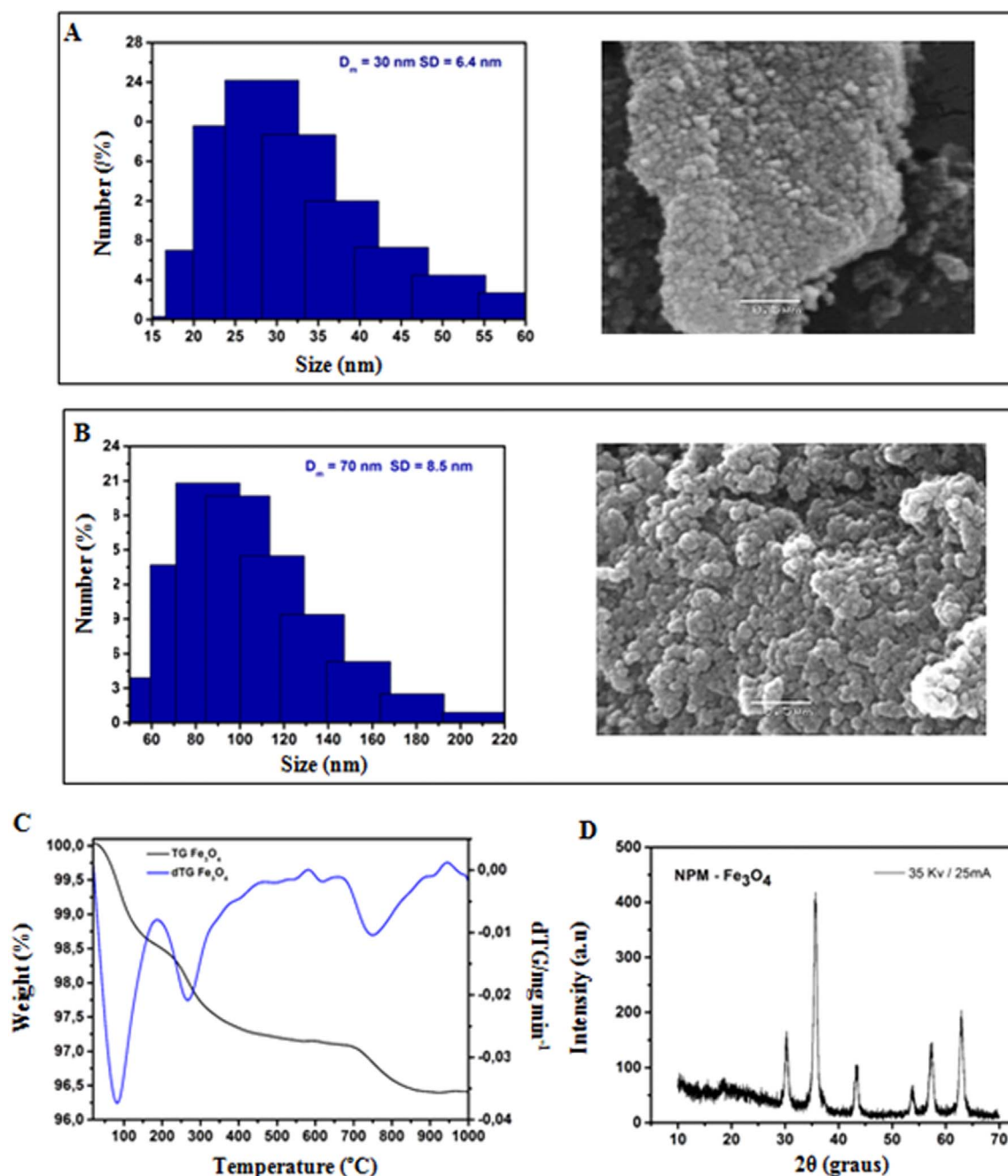


Fig. 2. TEM images and size distribution of Fe₃O₄ nanoparticles. (A) 30 nm, (B) 70 nm, (C) TG/dTG of Fe₃O₄ particles, and (D) XRD spectrum for Fe₃O₄.

3.3.4. Influence of pH

The pH value is an important parameter because influences the charge state of proteins, therefore influencing their interaction with MNPs [19]. To assess the effect of pH on protein depletion, a pH range from 5.5 to 8.5 of the MNPs/protein complex was evaluated. For this task, 5 µg of MNPs (70 nm) were added to 25 µL of serum diluted in 75 µL-Tris-HCl (pH 5.5 or 8.5) or PBS (pH 7.4) buffers. The samples were then shaken at 300 rpm for 30 min at 25 °C. According Fig. 3, panel D, lower adsorption capacity was obtained at pH 7.4, while highest values were observed at both pH 5.5 and 8.5. This observation may be explained by the maximum binding capacity between MNPs and proteins being reached at the pH near to the protein pI, as exemplified by albumin (pI of 5.2–5.9) and immunoglobulin (pI of 8.0–9.0), which together represent ca. 80% of the human serum. Such conditions greatly support the values observed in our study. Furthermore, such findings are in agreement with data previously

reported by Peng et al. [20], who found that the maximum adsorption of albumin on Fe₃O₄ MNPs is near to its pI. Therefore, considering that albumin is the most abundant protein in human serum, being the main interfering species in the detection of low abundance protein, and as our goal is to evaluate methods for removing abundant protein from serum samples, a pH value of 5.5 was selected to perform protein depletion using MNPs.

3.4. Protein depletion using Fe₃O₄ MNPs

Once the optimum conditions for the treatment of serum samples were found, the following procedure was applied to obtain the 1-DE profile of the supernatant, and to evaluate the removal of high abundance proteins using MNPs. It is important to emphasize here that no coating was applied to these nanoparticles, making their synthesis easier and being a procedure differential of this study. In

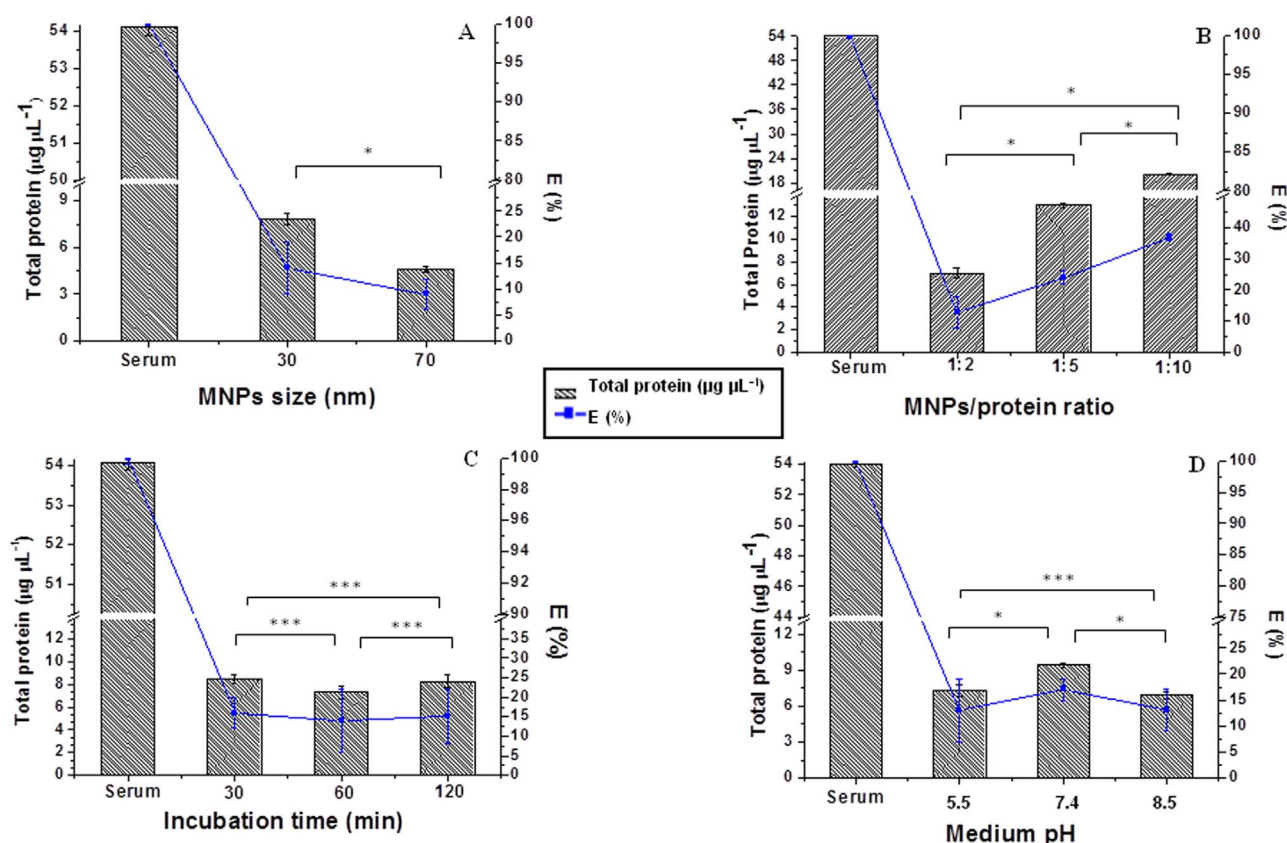


Fig. 3. Efficiency depletion for removing high abundant proteins. (A) MNPs size effect. (B) MNPs/protein ratio influence. (C) Influence of incubation time. (D) Effect of pH. Results obtained from three independent depletion experiments. The square curve represents the protein recovery (%) remaining after the depletion. Statistical analysis were performed using *t*-test and considered significant for * $p < 0.001$, and *** $p > 0.05$.

addition, this study has aim evaluated the proteins present in the supernatant after MNPs depletion procedure. Thus for protein depletion, 25 μL human serum diluted with 75 μL Tris-HCl pH 5.5 were shaken (300 rpm) with 5 μg MNPs (70 nm) for 30 min at 25 $^{\circ}\text{C}$. After this time, the supernatant and MNPs containing adsorbed proteins were separated magnetically. The supernatant was evaporated to dryness and reconstituted in buffer appropriate for 1-DE, and those proteins remaining in the supernatant were quantified. The quantification results showed the recoveries of supernatant proteins (ca 4.6 $\mu\text{g } \mu\text{L}^{-1}$) from ca. 54 $\mu\text{g } \mu\text{L}^{-1}$ of total protein in the serum sample (see bar graphic, Fig. 1, panel A).

Different proteins have different properties, which lead to different binding behaviors on MNPs. Although the underlying physical mechanism is not yet fully resolved, adsorption of proteins at the nanoparticle interface is a multifactorial process and not depends on the characteristics of the MNPs only, but also on the interacting proteins, medium, and several forces involved, *i.e.* van der Waals interactions, H-bonds, hydrophobic interactions, and, particularly, electrostatic interactions [19,21–24]. Fig. 1 (panel B) illustrates 1D-gel electrophoresis results, in which proteins present in the supernatant were incubated with MNPs under optimal conditions. When the MNP-based protein depletion is compared with the crude serum, an increase in the resolution of the bands is observed. Furthermore, the band resolution clearly indicates the depletion of some proteins of high molecular weight (> 80 kDa), as well as the enrichment of those of low molecular weight, when compared to crude serum. However, this method does not significantly deplete albumin, as highlighted by the intense band at ca. 66 kDa range (Fig. 1, panel B). This observation can be explained by larger proteins being likely to bind to MNPs more strongly than do smaller ones [15]. As albumin is a small protein, and there are large proteins with pI similar to that of albumin, the competition for adsorption sites on MNPs hampers albumin removal.

After electrophoresis separation, the bands were excised manually, subjected to enzymatic digestion, and analyzed through nanoLC-ESI-MS/MS. Seventy-two unique proteins were identified, as seen in Table 1, which provides a list of all proteins found in the supernatant. The MS/MS results confirmed the depletion of high molecular weight proteins (> 80 kDa), as well as the reduction of those high abundance proteins, such as transferrin and immunoglobulin, in the supernatant. Furthermore, proteins associated with important human disease, such as apolipoprotein B-100, gelsolin, and ceruloplasmin (all associated to cardiovascular disease) [25], fibronectin (ischemic stroke) [26], afamin (ovarian cancer) [27], and ficolin (infectious respiratory disease) [28] were identified using MNP depletion only, demonstrating the efficiency and reproducibility of this method in the enrichment of low abundance protein.

3.5. Sequential protein depletion using DTT and ACN

Recent studies have demonstrated that ACN and DTT can be used as efficient reagents for depleting abundant proteins in blood (serum and plasma) [10]. ACN demonstrated good efficiency in depleting proteins of high molecular weight from human plasma, and DTT in depleting those high abundant proteins rich in disulfide bonds, such as albumin and transferrin [29]. Based on such results, Fernández-Costa et al. [3] combined both chemical reagents sequentially, with the aim of removing high abundance proteins present in human serum. Based on that work, we evaluated this strategy (DTT+ACN) for simplifying the serum protein profile. Thus, human serum was firstly depleted with DTT, and the supernatant further treated with ACN, to efficiently reduce the dynamic range of proteins in the sample.

Incubation of a serum-DTT solution for 60 min at 37 $^{\circ}\text{C}$ resulted in a white solid precipitate. This observation may be explained by high temperature (37 $^{\circ}\text{C}$) strengthening the action of DTT by increasing the

Table 1
Identified protein after depletion using MNPs.

Protein full name	Entry name	pI	MM (Da)	Score	Coverage (%)
Apolipoprotein B-100	APOB_HUMAN	6.61	516,666	1346	23
Serum albumin	ALBU_HUMAN	5.92	71,317	2083	76
Histidine-rich glycoprotein	HRG_HUMAN	7.09	60,510	196	15
Alpha-2-macroglobulin	A2MG_HUMAN	6.00	164,614	1737	46
Complement C3	CO3_HUMAN	6.02	188,569	959	16
Protein AMBP	AMBP_HUMAN	5.95	39,886	76	17
Complement C4-A	CO4A_HUMAN	6.65	194,247	187	22
Ig gamma-1 chain C region	IGHG1_HUMAN	8.46	36,596	283	53
Apolipoprotein(a)	APOA_HUMAN	5.58	514,737	44	2
Ig mu chain C region	IGHM_HUMAN	6.35	49,960	433	36
Ig gamma-2 chain C region	IGHG2_HUMAN	7.66	36,505	121	26
Fibronectin	FINC_HUMAN	5.45	266,034	56	4
Complement factor H	CFAH_HUMAN	6.21	143,680	547	28
Ceruloplasmin	CERU_HUMAN	5.44	122,983	667	31
Ig lambda chain C regions	LAC_HUMAN	6.92	11,401	113	32
Serotransferrin	TRFE_HUMAN	6.81	79,280	1593	60
Ig kappa chain C region	IGKC_HUMAN	5.58	11,773	923	80
Complement C5	CO5_HUMAN	6.11	189,897	118	5
Plasminogen	PLMN_HUMAN	7.04	93,247	407	37
Complement factor B	CFAB_HUMAN	6.67	86,847	462	32
Plasma protease C1 inhibitor	IC1_HUMAN	6.09	55,347	310	24
Gelsolin	GELS_HUMAN	5.90	86,043	292	12
Complement C1s subcomponent	C1S_HUMAN	4.86	78,174	69	2
Antithrombin-III	ANT3_HUMAN	6.32	53,025	63	2
Complement component C7	CO7_HUMAN	6.09	96,650	45	2
Afamin	AFAM_HUMAN	5.64	70,963	50	15
Alpha-1B-glycoprotein	A1BG_HUMAN	5.58	54,809	169	23
Hemopexin	HEMO_HUMAN	6.55	52,385	416	45
Heparin cofactor 2	HEP2_HUMAN	6.41	57,205	63	15
C4b-binding protein alpha chain	C4BPA_HUMAN	7.15	69,042	149	8
Vitronectin	VTNC_HUMAN	5.55	55,069	61	5
N-acetylmuramoyl-L-alanine amidase	PGRP2_HUMAN	7.25	62,748	37	2
Angiotensinogen	ANGT_HUMAN	5.87	53,406	37	4
Ig alpha-1 chain C region	IGHA1_HUMAN	6.08	38,486	328	38
Alpha-1-antitrypsin	A1AT_HUMAN	5.37	46,878	1031	60
Ig gamma-3 chain C region	IGHG3_HUMAN	8.23	42,287	249	34
Alpha-2-HS-glycoprotein	FETUA_HUMAN	5.43	40,098	209	13
Kininogen-1	KNG1_HUMAN	6.34	72,996	143	11
Beta-2-glycoprotein 1	APOH_HUMAN	8.34	39,584	177	29
Ig gamma-4 chain C region	IGHG4_HUMAN	7.18	36,431	321	44
Ig heavy chain V-III region TIL	HV304_HUMAN	9.24	12,462	177	29
Alpha-1-acid glycoprotein 1	A1AG1_HUMAN	4.93	23,725	152	27
Ig heavy chain V-III region GAL	HV320_HUMAN	8.70	12,836	75	17
Ig heavy chain V-I region HG3	HV102_HUMAN	9.10	13,108	68	9
Haptoglobin-related protein	HPTR_HUMAN	6.63	39,518	62	3
Ig heavy chain V-II region ARH-77	HV209_HUMAN	8.45	16,389	49	6
Ig heavy chain V-III region HIL	HV310_HUMAN	9.43	13,671	48	13
Haptoglobin	HPT_HUMAN	6.13	45,861	198	25
Clusterin	CLUS_HUMAN	5.89	53,031	184	21
Apolipoprotein E	APOE_HUMAN	5.65	36,246	365	56
Ficolin-3	FCN3_HUMAN	6.20	33,395	60	6
Apolipoprotein L1	APOL1_HUMAN	5.60	44,004	54	7
Apolipoprotein D	APOD_HUMAN	5.06	21,547	162	28
Complement C1q subcomponent subunit B	C1QB_HUMAN	8.83	26,670	247	20
Serum amyloid P-component	SAMP_HUMAN	6.10	25,485	165	27
Ig lambda chain V-III region LOI	LV302_HUMAN	4.95	12,042	59	21
Ig lambda chain V-I region HA	LV102_HUMAN	9.07	12,003	77	11
Apolipoprotein A-I	APOA1_HUMAN	5.56	30,759	939	66
Ig kappa chain V-III region SIE	KV302_HUMAN	8.70	11,882	278	39
Ig kappa chain V-I region AG	KV101_HUMAN	5.67	12,099	302	31
Ig kappa chain V-I region EU	KV106_HUMAN	8.62	11,895	122	26
Ig kappa chain V-II region TEW	KV204_HUMAN	5.69	12,422	168	35
Ig kappa chain V-IV region Len	KV402_HUMAN	7.92	12,746	84	36
Ig kappa chain V-III region VG (Fragment)	KV309_HUMAN	4.85	12,681	63	7
Ig kappa chain V-I region Lay	KV113_HUMAN	7.96	11,941	52	8
Ig lambda chain V-I region NIG-64	LV104_HUMAN	4.66	11,561	48	7
Ig kappa chain V-I region CAR	KV104_HUMAN	9.47	11,810	148	24
Immunoglobulin J chain	IGJ_HUMAN	5.12	18,543	49	7
Ig kappa chain V-IV region	KV401_HUMAN	5.09	13,486	48	7
Complement C1q subcomponent subunit C	C1QC_HUMAN	8.61	25,985	47	4
Transthyretin	TTHY_HUMAN	5.52	15,991	330	53
Serum amyloid A-4 protein	SAA4_HUMAN	9.27	14,854	66	15

reaction yields, then disrupting intermolecular and intramolecular disulfide bonds of proteins. Such disruption promotes denaturation of thiol-rich proteins, allowing subsequent precipitation. After precipitation, both supernatant and pellet were separated by centrifugation (13,000g), and the supernatant then diluted with water (45 μL) and vortexed to mix. After adding the ACN dropwise (85 μL – see Section 2), the solution became cloudy as the aggregation of proteins started. This aggregation may be caused by organic solvent lowering the dielectric constant of the protein solution, displacing the water molecules around the hydrophilic regions on the protein surface, enhancing the electrostatic attractions among charged protein molecules and minimizing the hydrophobic interactions between proteins. Therefore, the proteins are then aggregated and precipitated. Prior to electrophoresis analysis, protein quantification was performed, and it was found that after depletion only 3.6 $\mu\text{g } \mu\text{L}^{-1}$ (bar graphic, Fig. 1, panel A) of the original protein content was preserved. Because of the low mass of resulting protein in the supernatant, in order to have sufficient protein, new depletions were performed, which were pooled for gel electrophoresis separation. The 1-DE profile obtained after the sequential depletion (see Fig. 1, panel B) shows a decrease in high molecular weight proteins (characterized by ACN depletion) and also the enrichment of some proteins (characterized by DDT depletion), the presence of which are masked by albumin in the crude sample. Such results are in accord with those reported in the literature [3,10].

Table 2 shows the proteins identified in serum after the depletion process, consisting of 20 different proteins identified using nanoLC-ESI-MS/MS. The result is consistent with 1-D gel electrophoretic analysis, which indicated the presence of immunoglobulin light chain (e.g. Ig kappa chain C region), and the enrichment of either apolipoproteins (A-I, A-II, A-IV, C-III and D) or others proteins of low molecular weight, such as alpha-1-antitrypsin (~46 kDa) and haptoglobin (~45 kDa). Furthermore, glycosylated proteins were also identified using this depletion method: for example alpha-1-acid-glycoprotein 1, alpha-2-HS-glycoprotein, and zinc-alpha-2-glycoprotein. All these glycoproteins have been associated as candidate biomarkers for diagnosis of important human diseases. For example, Yin et al. [25] report alpha-1-acid-glycoprotein 1 as one of the four proteins associated with risk of atherosclerotic cardiovascular disease. Fan et al. [30] and Jung et al. [31] suggest the involvement of alpha-2-HS-glycoprotein and zinc-alpha-2-glycoprotein in the pathogenesis of colorectal cancer and prostate cancer, respectively.

3.6. Protein equalization using PM

PM is a commercial apparatus developed by BioRad that facilitates a reduction in the dynamic range of proteins in complex biological samples such as serum. It is based on the selective adsorption of proteins on a chromatographic support containing a library of hexapeptides [32]. Proteins of low and high abundance are enriched and reduced respectively, enabling detection of the low abundance proteins, easing the identification of biomarkers. Human serum (200 μL) was applied to the column and incubated for 2 h. After washing the column with buffer, the elution was performed three times, for 15 min, using 100 μL of specific solution (see Section 2.5.3). The eluted fractions were pooled and subsequently quantified. According to total protein quantification, ca. 3.3 $\mu\text{g } \mu\text{L}^{-1}$ (n=3) of protein were recovered (bar graphic, Fig. 1, panel A). Because of the low protein mass recovery, new depletions were needed. About 30 μg were used to perform 1-DE separation (Fig. 1, panel B), being possible observing lower density of the gel bands, confirming a decrease in the dynamic range of proteins in the sample. MS analysis allowed the identification of 30 unique proteins (Table 3) from gel bands previously excised and digested. Proteins such as fibulin-1 (ca. 80 kDa), component complement C9 (ca. 64 kDa) and complement C1q subcomponent subunit C (ca. 25 kDa) associated with breast cancer [33], inflammatory [34] and immunological [35] diseases respectively, were found. Other important proteins associated with human disease, such as apolipoproteins, were also identified using this approach.

3.7. Comparison of depletion methods

The efficiency and reproducibility of the depletion methods were evaluated by a duplicate analysis of the protein content of supernatant on 1-DE gels. No significant differences between gels duplicates were observed (Supplementary material, Fig. 1S).

Not surprisingly, among the depletion methods evaluated, the most efficient depletion of high abundance proteins was achieved using PM equalization, which resulted in the lowest protein content, $3.3 \pm 0.2 \mu\text{g } \mu\text{L}^{-1}$ (n=3) in the supernatant, while the use of DTT/ACN resulted in a supernatant with a protein content of $3.6 \pm 0.1 \mu\text{g } \mu\text{L}^{-1}$ (n=3). The MNP depletion rendered a supernatant with a total protein content of $4.6 \pm 0.2 \mu\text{g } \mu\text{L}^{-1}$ (n=3). Five, six, and nine proteins, which are included in the group of the most abundant ones (top 10) [36], were identified after depletion with DTT/ACN, PM and MNP methods, respectively. This result suggests that DTT/ACN is as efficient as PM for

Table 2
Identified protein after depletion using DTT/ACN.

Protein full name	Entry name	pI	MM (Da)	Score	% coverage
Ig kappa chain C region	IGKC_HUMAN	5.58	11,773	33	18
Apolipoprotein A-I	APOA1_HUMAN	5.56	30,759	718	67
Serotransferrin	TRFE_HUMAN	6.81	79,280	47	5
Serum albumin	ALBU_HUMAN	5.92	71,317	151	21
Alpha-2-HS-glycoprotein	FETUA_HUMAN	5.43	40,098	171	15
Angiotensinogen	ANGT_HUMAN	5.87	534,069	126	8
Alpha-1-antitrypsin	A1AT_HUMAN	5.37	46,878	824	34
Alpha-1-acid glycoprotein 1	A1AG1_HUMAN	4.93	23,725	167	27
Putative zinc-alpha-2-glycoprotein-like 1	ZAGL1_HUMAN	5.84	23,080	67	4
Zinc-alpha-2-glycoprotein	ZA2G_HUMAN	5.57	34,079	154	17
Apolipoprotein A-IV	APOA4_HUMAN	5.28	45,371	499	38
Haptoglobin	HPT_HUMAN	6.13	45,861	104	5
Alpha-1-acid glycoprotein 2	A1AG2_HUMAN	5.03	23,873	79	17
Haptoglobin-related protein	HPTR_HUMAN	6.63	39,518	202	9
Apolipoprotein C-III	APOC3_HUMAN	5.23	10,845	111	23
- Apolipoprotein A-II	APOA2_HUMAN	6.26	11,282	92	21
Hemoglobin subunit beta	HBB_HUMAN	6.75	16,102	162	52
Hemoglobin subunit alpha	HBA_HUMAN	8.72	15,305	146	44
Apolipoprotein D	APOD_HUMAN	5.06	21,547	100	17
Transthyretin	TTHY_HUMAN	5.52	15,991	101	13

Table 3
Identified protein after PM-equalization.

Protein full name	Entry name	pI	MM (Da)	Score	%coverage
Complement factor H	CFAH_HUMAN	6.21	143,680	286	20
Vitronectin	VTNC_HUMAN	5.55	55,069	231	24
Complement C4-A	CO4A_HUMAN	6.65	194,247	627	13
Complement C3	CO3_HUMAN	6.02	188,569	335	12
Antithrombin-III	ANT3_HUMAN	6.32	53,025	523	38
Fibulin-1	FBLN1_HUMAN	5.07	81,268	116	8
Ig mu chain C region	IGHM_HUMAN	6.35	49,960	301	26
C4b-binding protein alpha chain	C4BPA_HUMAN	7.15	69,042	208	14
Plasminogen	PLMN_HUMAN	7.04	93,247	41	5
Serum albumin	ALBU_HUMAN	5.92	71,317	170	21
Coiled-coil domain-containing protein 34	CCD34_HUMAN	7.07	43,598	34	3
Complement component C9	CO9_HUMAN	5.43	64,615	60	8
Lipopolysaccharide-binding protein	LBP_HUMAN	6.23	53,521	43	8
Ig alpha-1 chain C region	IGHA1_HUMAN	6.08	38,486	70	5
Ig gamma-1 chain C region	IGHG1_HUMAN	8.46	36,596	74	11
Alpha-1-antitrypsin	A1AT_HUMAN	5.37	46,878	41	10
Clusterin	CLUS_HUMAN	5.89	53,031	80	14
Apolipoprotein A-IV	APOA4_HUMAN	5.28	45,371	59	14
Complement factor H-related protein 1	CFHR1_HUMAN	7.38	38,766	42	6
Prothrombin	THRB_HUMAN	5.64	71,475	194	14
Apolipoprotein E	APOE_HUMAN	5.65	36,246	230	32
Ficolin-3	FCN3_HUMAN	6.20	33,395	31	6
Serum amyloid P-component	SAMP_HUMAN	6.10	25,485	101	20
Ig kappa chain C region	IGKC_HUMAN	5.58	11,773	47	18
Ig kappa chain V-II region Cum	KV201_HUMAN	5.28	12,782	39	11
Apolipoprotein D	APOD_HUMAN	5.06	21,547	33	12
Complement C1q subcomponent subunit B	C1QB_HUMAN	8.83	26,670	28	5
Complement C1q subcomponent subunit C	C1QC_HUMAN	8.61	25,774	49	10
Apolipoprotein A-I	APOA1_HUMAN	5.56	30,759	48	23
Immunoglobulin J chain	IGJ_HUMAN	5.12	18,543	31	7

the removal of such proteins from the serum. On the other hand, the MNP method presents the highest dynamic range of serum protein concentrations, detecting almost all the proteins in the top 10. In addition, the number of low abundance proteins identified is higher in the MNP procedure (Table 1) than those obtained with DTT/ACN (Table 2), or PM (Table 3) procedures, suggesting that MNP depletion may be an effective strategy for low abundance protein enrichment in the search for candidates for human disease biomarkers. This observation may be explained by the albumin sponge effect [36]. In other words, as albumin has the capacity of binding to certain low molecular weight protein, and as albumin is not fully depleted from serum when the MNPs depletion method is used, many of these low molecular weight proteins may be detected in the serum after MNPs depletion. Another point to note is that PM, although decreasing the dynamic range of protein levels in complex biological samples, presents some limitations compared to MNPs and DTT/ACN depletion, such as its slowness, cost, and the large amount of sample needed for processing. In addition, DTT/ACN depletion shows similar 1-DE gel profiles to those obtained through PM equalization. In this regard, ACN/DTT or MNP depletions are welcome alternatives to immunoaffinity depletion.

A total of 122 proteins were identified using GeLC-MS/MS over the three depletion methods evaluated. Fig. 4 (panel E) shows the overlap of these identified proteins, which were functionally categorized based on universal Gene Ontology (GO) annotations terms using Universal Protein Resource (UniProt) software (www.uniprot.org). As shown in the Fig. 4 (panels A, B, C, and D), and in Supplementary material (Tables 4–6S), the proteins were linked to at least one annotation term within the GO categories: Molecular Function, Biological Process, Cellular Component, and Disease Involvement. In the Molecular Function category, several GO terms were observed, however five main groups were highlighted: (i) signaling pathway activity, (ii) enzyme regulator, (iii) molecular transporter, (iv) molecular interaction, and (v) function not determined. The molecular interaction function was the category which represented the largest number of proteins, followed by signaling pathway activity and

molecular transporter (see Fig. 4, panel A). Fig. 4, panel B, shows those proteins overrepresented or underrepresented in the Biological Process category. Immune and inflammatory responses subcategories were the biological processes with the highest number of proteins. This finding is reasonable given that serum contains many proteins involved in immune response, such as immunoglobulin family, apolipoprotein family, and complement component family proteins. In the Cellular Component category (Fig. 4, panel C), five groups were subcategorized: (i) extracellular, (ii) cell surface, (iii) plasma membrane, (iv) cytoplasm, and (v) nucleus. As observed in Fig. 4, panel C, extracellular proteins and plasma membrane proteins were overrepresented, while nucleus proteins, cell surface proteins, and cytoplasm proteins were found to be underrepresented. Regarding the disease involvement category, a diversification of the identified proteins associated with different diseases was observed (Fig. 4, panel D). For this study, five groups were categorized: (i) metabolic disease, (ii) neurological disease, (iii) immunological disease, (iv) inflammatory disease, and (v) others. In contrast to panel B, Fig. 4, the immunological disease and inflammatory disease subcategories were observed to be underrepresented. This fact is justified because the UniProt program presents proteins associated with disease biomarkers only and at least with one publication reported in literature. In other words, although some proteins are involved in biological processes related to immune response (Fig. 4, panel B), there are few proteins reported as biomarkers for inflammatory and immunological diseases (Fig. 4, panel D).

4. Conclusions

The main purpose of this work was broadly achieved. Our results clearly confirm the potential of the strategies evaluated for the depletion of major serum proteins, in order to facilitate low abundance protein detection. MNP depletion proved to be a versatile and reproducible sample preparation technique, being efficient in removing higher molecular weight proteins (> 80 kDa), as well as in enriching

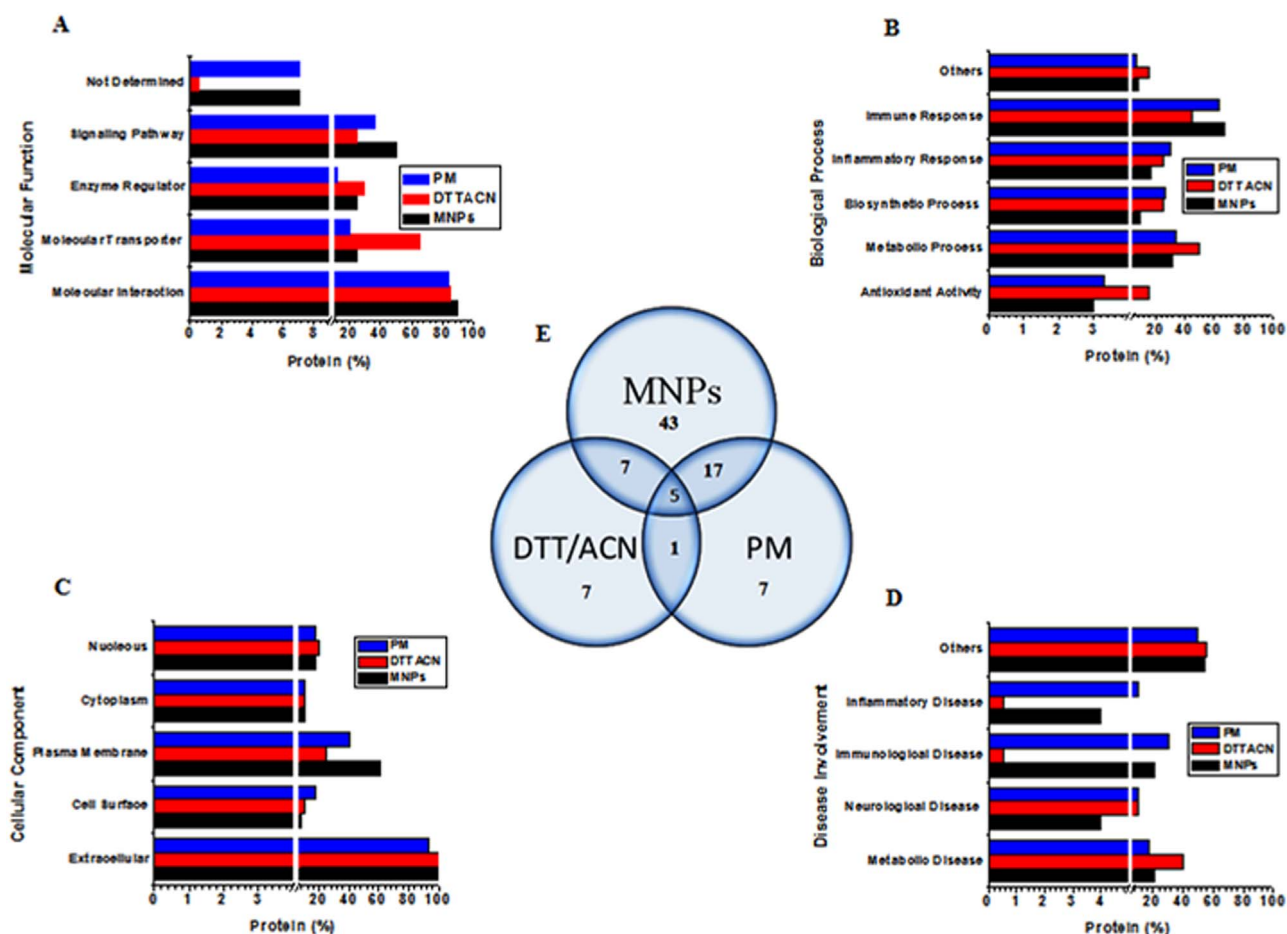


Fig. 4. Proteomic panel comparing the depletion methods. Bars show the percentage of proteins with the indicated Gene Ontology annotations for: (A) Molecular Function; (B) Biological Process; (C) Cellular Component; (D) Disease Involvement. (E) Venn diagram shows the overlap of protein identified through three depletion methods (MNPs, DTT/ACN, and PM). Gene ontology annotations were based on UNIPROT program.

those of low molecular weight, such as gelsolin and afamin, which are associated with important human disease. In fact, such strategy presented the highest number of identified proteins. Results obtained by the sequential DTT/ACN method showed not only a decrease in high molecular weight proteins (characterized by ACN depletion), but also the enrichment of some proteins (characterized by DDT depletion) currently masked by albumin in the crude sample. However, this approach presented a reduction in the number of identified proteins, representing a negative point of the strategy. Regarding PM equalization, there is no doubt about the efficiency of this strategy in decreasing the dynamic range of protein levels in complex biological samples, such as human serum. Nevertheless, when PM is compared with MNP or DTT/ACN depletion, it presents some limitations, such as its cost, and the large amount of sample needed for electrophoretic analysis. Furthermore, PM equalization gel profile (1-DE) showed some similarities to those obtained using DTT/ACN depletion. Thus, based on the results obtained, MNP depletion mainly, or DTT/ACN depletion secondly, can be considered as powerful alternatives to the immunochemistry-based protein depletion techniques currently used for the removal of high abundance proteins from serum prior to MS-based proteomic analyses, opening up their application in the search for biomarkers in clinical samples.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.talanta.2017.03.091](https://doi.org/10.1016/j.talanta.2017.03.091).

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